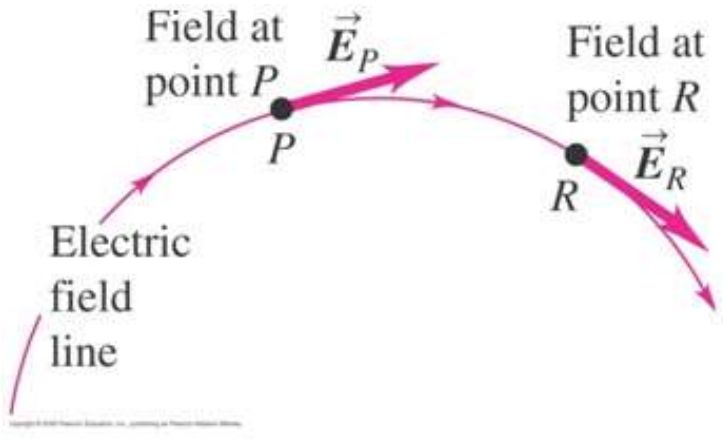
# Gauss's Law and applications

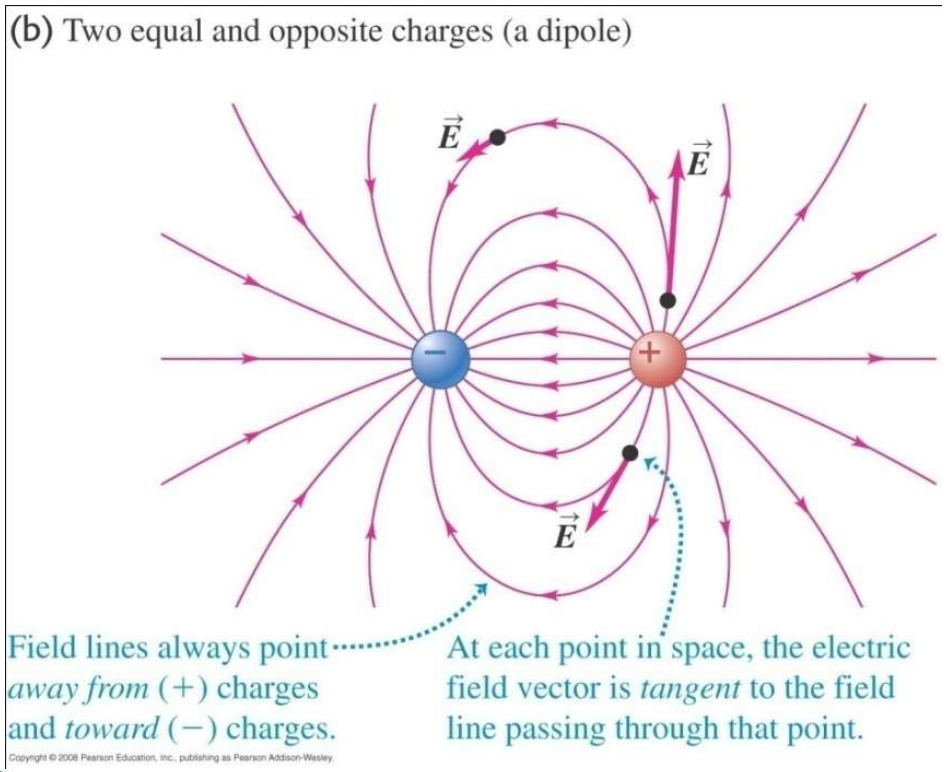
# Electric field

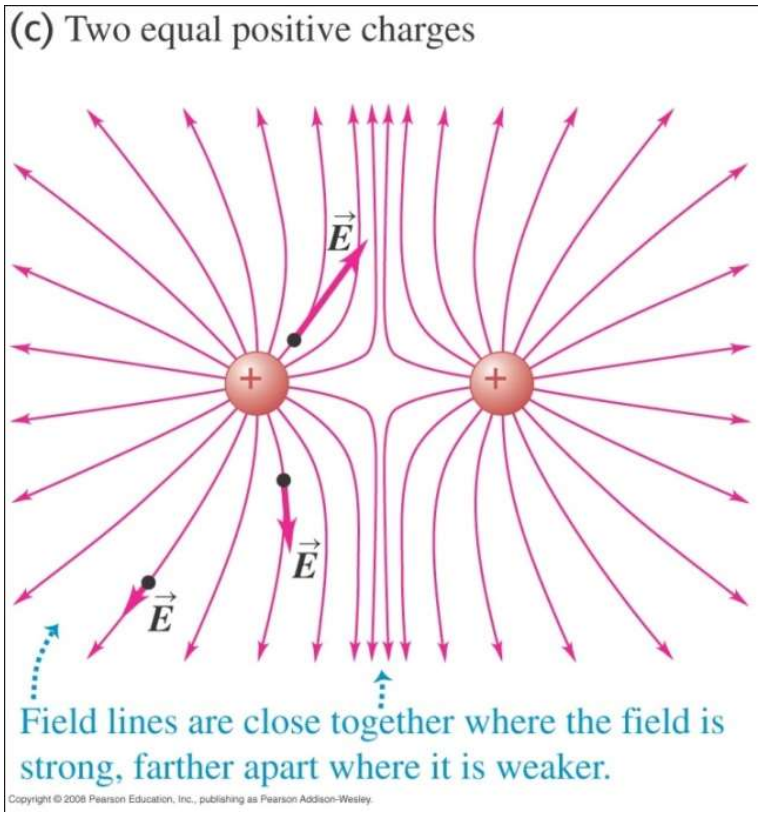
- An **electric field** is a region where an electric charge experiences an electric force.
- Electric fields are represented by field lines.
- The tangent at any point on a field line is the direction of the force that a positive charge would experience when placed at that point.



# Electric field lines of two nearby point charges

- The electric field of two nearby point charges is the resultant of the fields of the two charges.



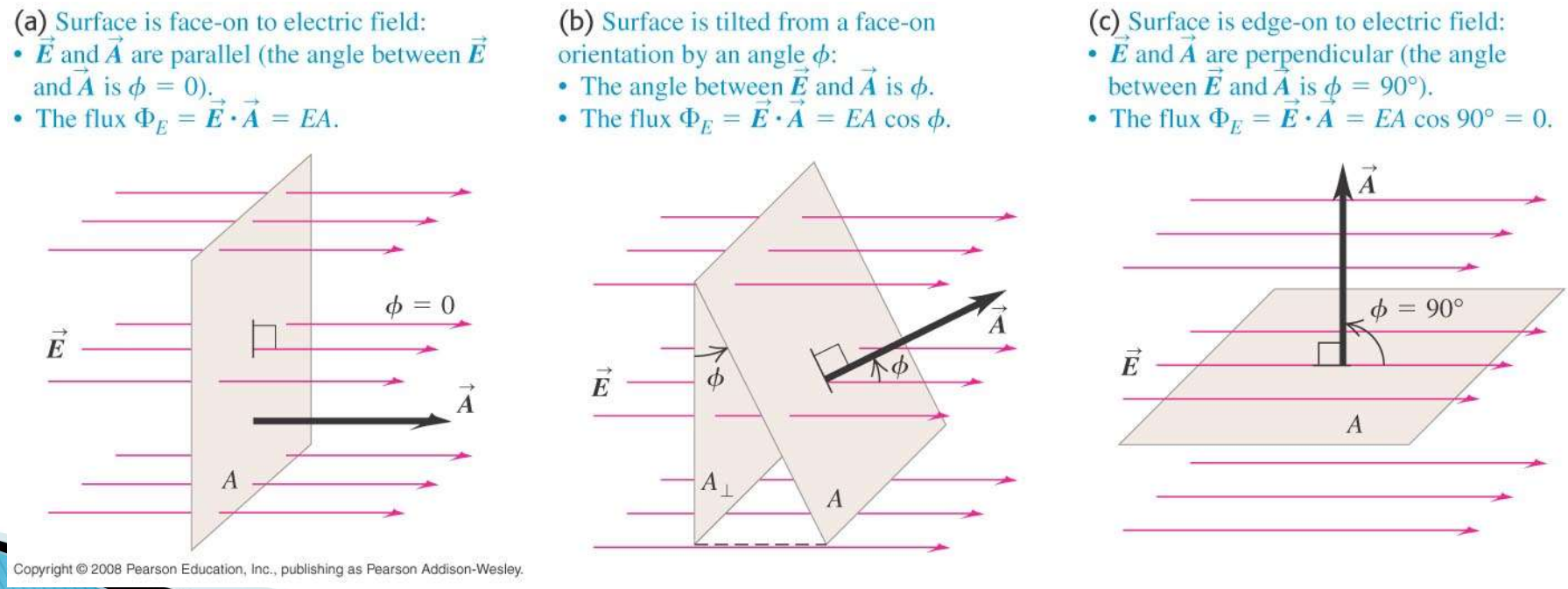



# Electric flux

- **Electric flux,** $\Phi$ is defined as $\Phi = \vec{E} \cdot \vec{A} = EA\cos\phi$

  where $\phi$ is the angle between the normal to the surface and the direction of $\vec{E}$ .

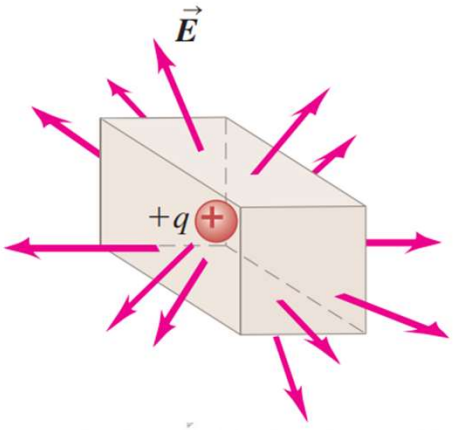
- If $\vec{E}$ is non-uniform, we calculate the electric flux as $\Phi = \int \vec{E} \cdot d\vec{A}$




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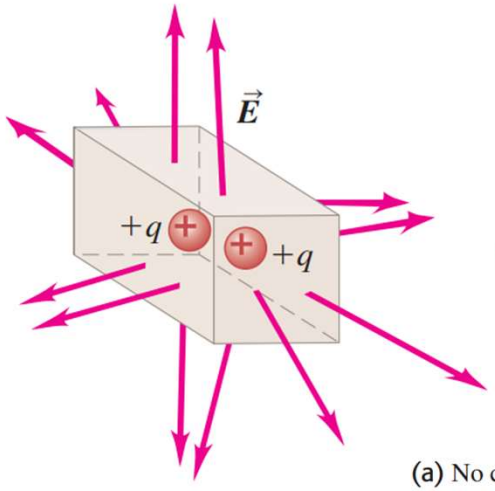
# Electric flux for different scenarios inside a rectangular box

(a) Positive charge inside box, outward flux

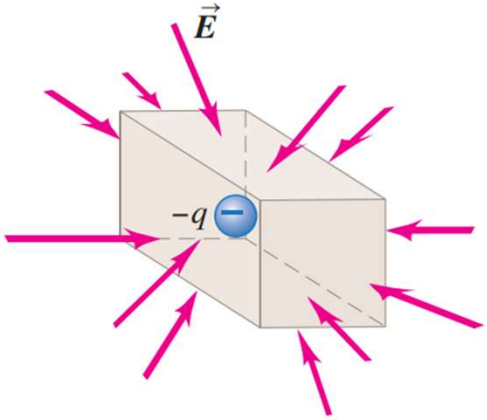



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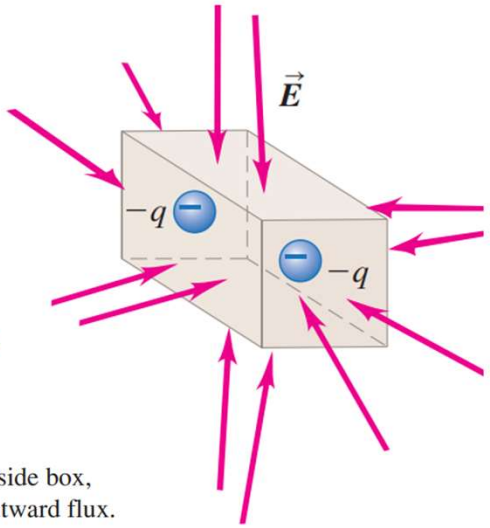
(b) Positive charges inside box, outward flux



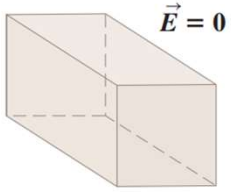

(c) Negative charge inside box, inward flux



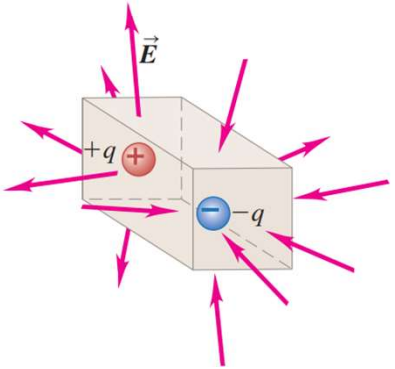

(d) Negative charges inside box, inward flux




(a) No charge inside box, zero flux

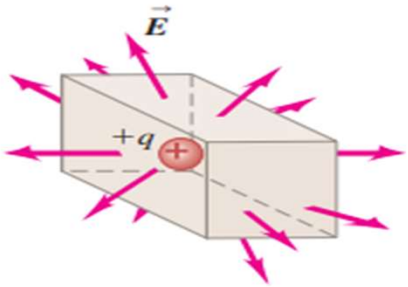



(b) Zero *net* charge inside box, inward flux cancels outward flux.

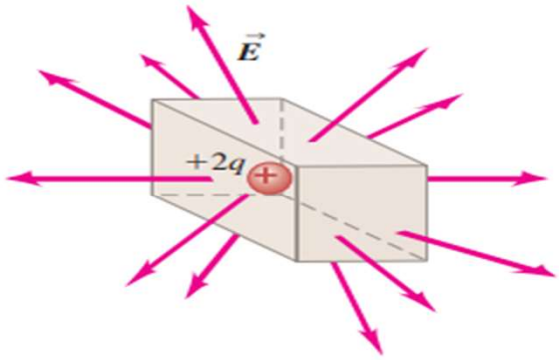


# Effect of Electric flux for different scenarios inside a rectangular box

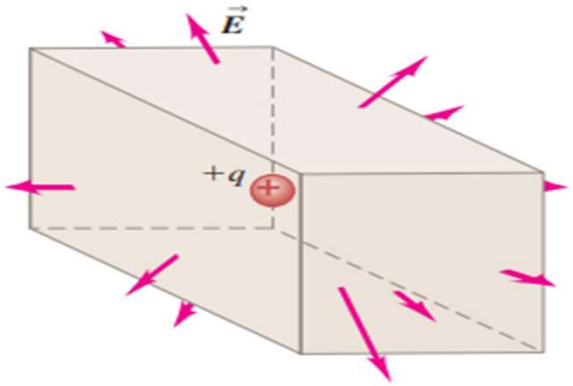
(a) A box containing a charge




(b) Doubling the enclosed charge doubles the flux.




(c) Doubling the box dimensions *does not change* the flux.




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# Summary from the scenarios

1. Whether there is a net outward or inward electric flux through a closed surface depends on the sign of the enclosed charge.
2. Charges *outside* the surface do not give a net electric flux through the surface.
3. The net electric flux is directly proportional to the net amount of charge enclosed within the surface but is otherwise independent of the size of the closed surface.

- Apart from rectangular shaped surfaces, the above results hold even for other closed surfaces.
- The above also are qualitative statement of Gauss's law.

# Example

A disk of radius 0.10 m is oriented with its normal unit vector $\hat{\boldsymbol{n}}$ at 30° to a uniform electric field $\vec{\boldsymbol{E}}$ of magnitude $2.0 \times 10^3$ N/C

(a) What is the Electric flux through the disk? (b) What is the flux when E is perpendicular to ***n***?(c) What is the flux when E is parallel to ***n***?

(a) The area is $A = \pi(0.10\ \text{m})^2 = 0.0314\ \text{m}^2$ and the angle between $\vec{\boldsymbol{E}}$ and $\vec{\boldsymbol{A}} = A\hat{\boldsymbol{n}}$ is $\phi = 30°$

$$\Phi_E = EA\cos\phi = (2.0 \times 10^3\ \text{N/C})(0.0314\ \text{m}^2)(\cos 30°)$$
$$= 54\ \text{N}\cdot\text{m}^2/\text{C}$$

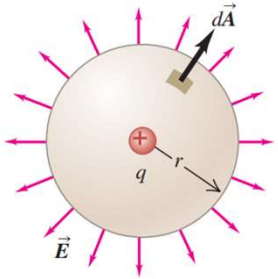
(b) The normal to the disk is now perpendicular to $\vec{\boldsymbol{E}}$, so $\phi = 90°$, $\cos\phi = 0$, and $\Phi_E = 0$.

(c) The normal to the disk is parallel to $\vec{E}$, so $\phi = 0$ and $\cos\phi = 1$:

$$\Phi_E = EA\cos\phi = (2.0 \times 10^3\ \text{N/C})(0.0314\ \text{m}^2)(1)$$
$$= 63\ \text{N}\cdot\text{m}^2/\text{C}$$

# Example

A point charge $q = +3.0\ \mu\text{C}$ is surrounded by an imaginary sphere of radius $r = 0.20$ m centered on the charge (see fig). Find the resulting electric flux through the sphere.




$\Phi_E = \int E\, dA$. At any point on the sphere of radius $r$ the electric field has the same magnitude $E = q/4\pi\epsilon_0 r^2$. Hence $E$ can be taken outside the integral, which becomes $\Phi_E = E\int dA = EA$.

$$\Phi_E = EA = \frac{q}{4\pi\epsilon_0 r^2}\,4\pi r^2 = \frac{q}{\epsilon_0}$$

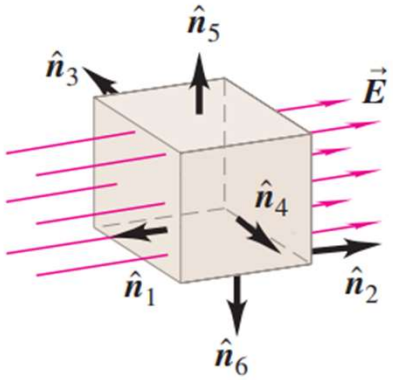
$$= \frac{3.0 \times 10^{-6}\ \text{C}}{8.85 \times 10^{-12}\ \text{C}^2/\text{N}\cdot\text{m}^2} = 3.4 \times 10^5\ \text{N}\cdot\text{m}^2/\text{C}$$

The radius $r$ of the sphere cancels out of the result for $\Phi_E$. We would have obtained the same flux with a sphere of radius 2.0 m or 200 m.

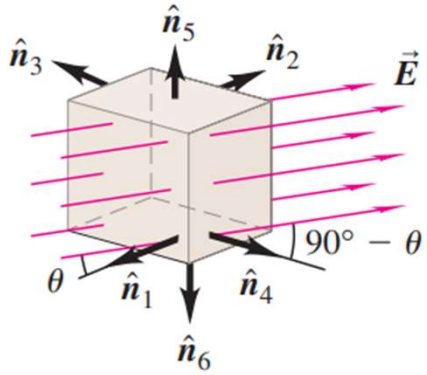
# Example

An imaginary cubical surface of side $L$ is in a region of uniform electric field $\vec{E}$. Find the electric flux through each face of the cube and the total flux through the cube when (a) it is oriented with two of its faces perpendicular to $\vec{E}$ (see fig) and (b) the cube is turned by an angle $\theta$ about a vertical axis.

(a)




(b)



# Symmetry

- Many times symmetry simplifies problems.
- Many physical systems have symmetry.
- If rotate a cylinder along its axis, it does not change.
- A sphere looks the same if it is rotated along an axis that passes through its centre.
- Other examples of symmetry can be found in the below: http://privatekgsmaths.weebly.com/symmetry-s1-notes.html

# Gauss' law

- Gauss' law states that "The **net electric flux** coming out of a closed surface is proportional to the net electric charge inside the surface."
- Mathematically,

$$\Phi = \int \vec{E} \cdot d\vec{A} \propto q$$

or

$$\Phi = \frac{q}{\varepsilon}$$

- If the charge is in vacuum, then $\Phi = \frac{q}{\varepsilon_o}$

# Gauss's law

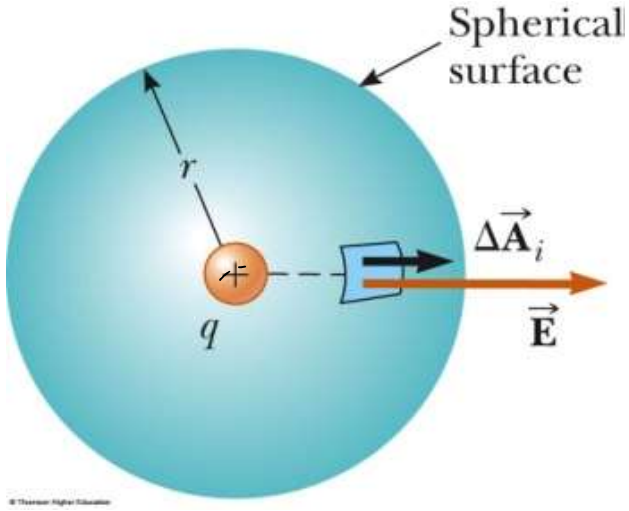
- Gauss's law relates electric field at points on a closed surface to the ***net charge enclosed*** by the surface.
- It helps to calculate electric fields using symmetry considerations.
- It helps us understand how electric charge distributes itself over conducting surfaces.
- It helps us gain a deeper insight into electric fields.

# Point charge inside a Spherical surface

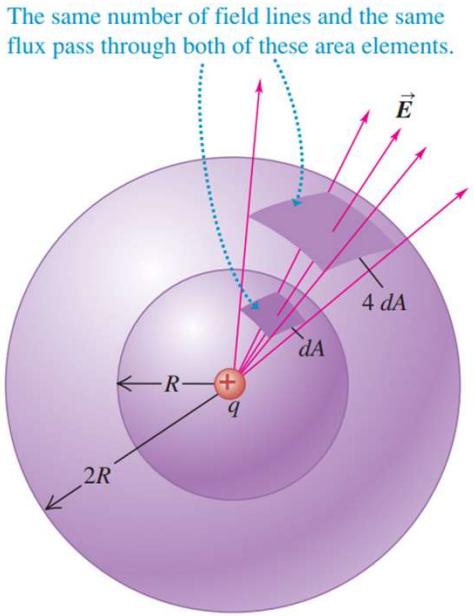
- Suppose a positive point charge, q, is at the center of a sphere radius r.

$$\phi = \int \vec{E} \cdot d\vec{A} = \int E\, dA \cos\theta = \int E\, dA$$

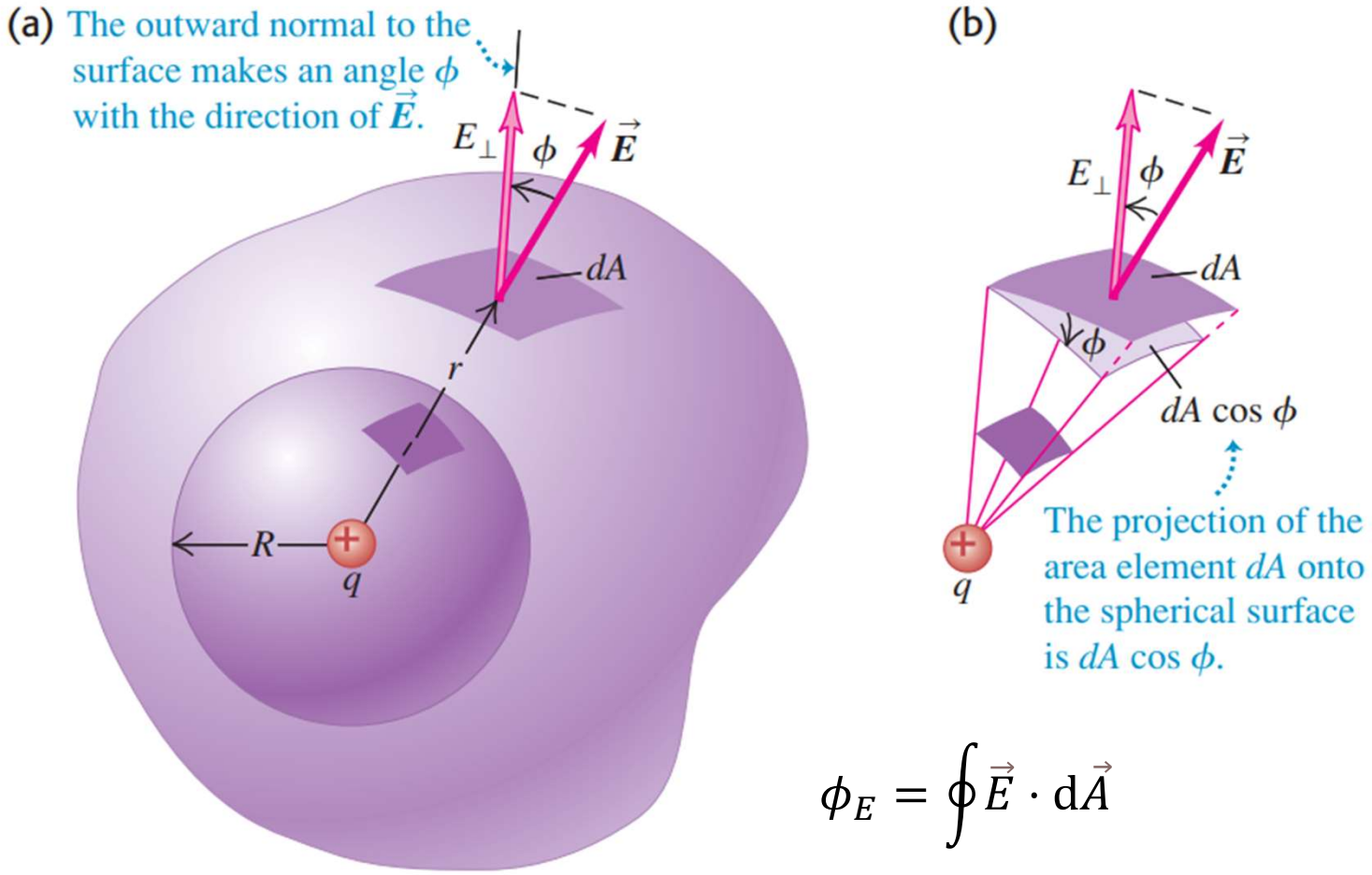
$$\phi = E \int dA = EA = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2} \times 4\pi r^2$$




- $\phi = \frac{q}{\varepsilon_0}$
- The flux is independent of the radius *r* and depends only on the charge enclosed by the sphere.
- If the radius is doubled, then also the flux remain same!
- If the charge is negative, the flux is negative.



# Point charge inside a non-spherical surface

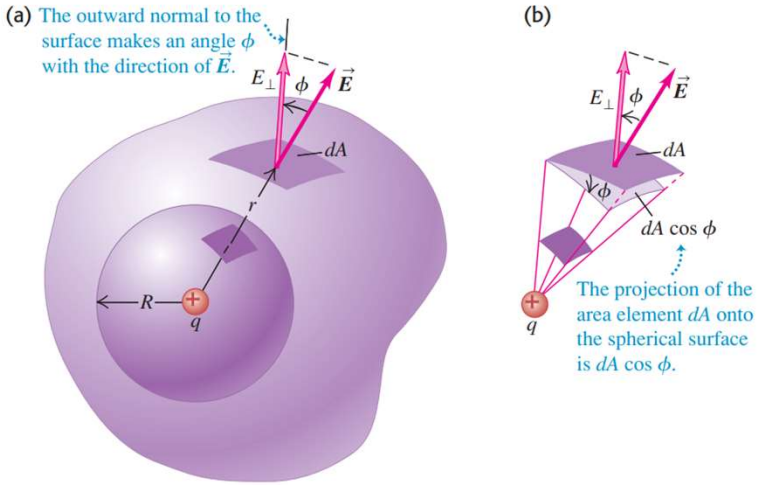



$$\phi_E = \oint \vec{E} \cdot \mathrm{d}\vec{A}$$

- The circle on the integral reminds us that the integration is on a closed surface.

# Point charge inside a non-spherical surface

- The electric flux  through the irregular surface should be the same as that through the sphere.



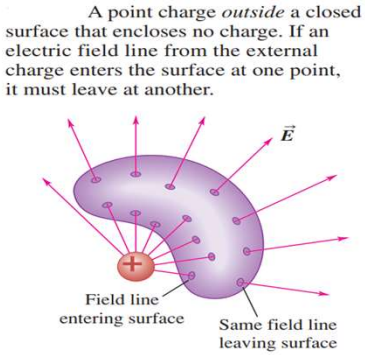

- Therefore, the electric flux through the irregular non-spherical surface is

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{q}{\epsilon_0}$$

# Charge outside a surface

- If there is no charge enclosed by a surface or if a charge is outside an enclosed surface, then

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = 0$$




- If the charge is outside an enclosed surface, the flux is zero even though the electric field is non-zero.

This is the mathematical statement that when a region contains no charge, any field lines caused by charges *outside* the region that enter on one side must leave again on the other side.

*Electric field lines can begin or end inside a region of space only when there is charge in that region.*

# General form of Gauss's law

- If there are charges $q_1$, $q_2$ .. etc enclosed by a surface, then the total charge enclosed is $Q_{enc} = q_1 + q_2 + \ldots$

- We need to calculate the net Electric field due to the all the charges, so that

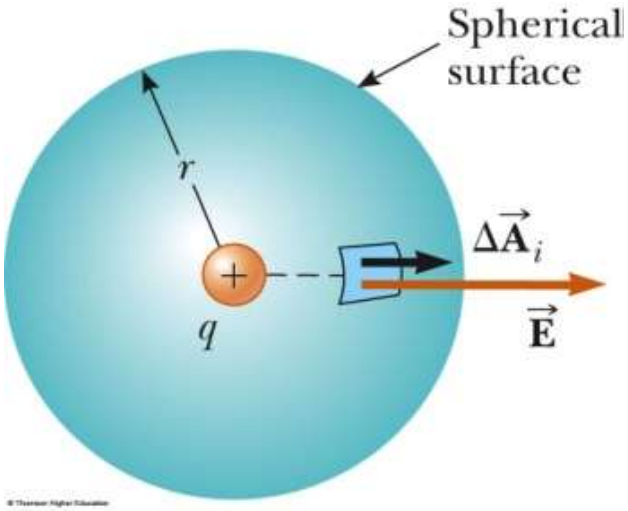
$$\Phi_E = \oint E\cos\phi\, dA = \oint E_\perp\, dA = \oint \vec{E}\cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0} \quad \text{(various forms of Gauss's law)}$$

**The total electric flux through a closed surface is equal to the total (net) electric charge inside the surface, divided by $\epsilon_0$.**

**CAUTION** **Gaussian surfaces are imaginary** Remember that the closed surface in Gauss's law is *imaginary;* there need not be any material object at the position of the surface. We often refer to a closed surface used in Gauss's law as a **Gaussian surface.**

# Coloumb's law from Gauss' law

- Suppose a positive point charge, q, is at the center of a sphere radius *r.*




Total flux $\Phi = \int \vec{E} \cdot d\vec{A}$

$= \int E dA$ E and A are parallel

$= E \int dA$ E is constant all over A

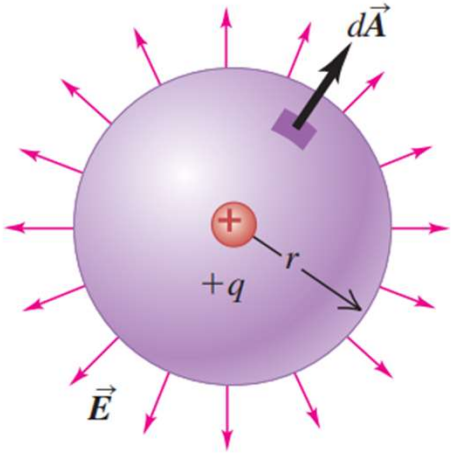
$= 4\pi r^2 E = \frac{q}{\varepsilon_o}$

Hence $E = \frac{q}{4\pi\varepsilon_o r^2}$

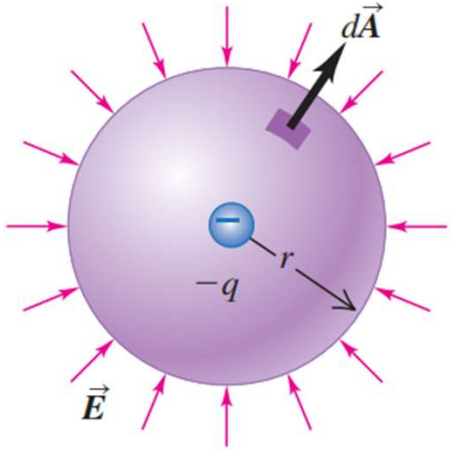
This formula is the same as the one derived using Coulomb's law.

# Direction of Electric field and sign of Electric flux

(a) Gaussian surface around positive charge: positive (outward) flux



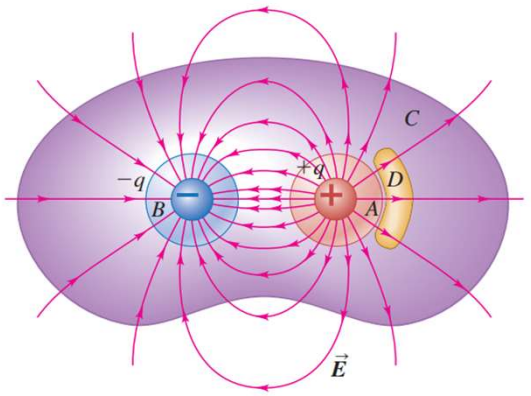

(b) Gaussian surface around negative charge: negative (inward) flux




$$\Phi_E = \oint E\cos\phi\, dA = \oint E_\perp\, dA = \oint \vec{E}\cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0}$$ (various forms of Gauss's law)

# Example

What is the electric flux through each of the closed surfaces A, B, C and D as shown in the below figure which shows a dipole?




- Apply Gauss's law to A, we get $\phi_A = \frac{q}{\varepsilon_0}$
- Apply Gauss's law to B, we get $\phi_B = -\frac{q}{\varepsilon_0}$
- Apply Gauss's law to C and D, we get the flux to be zero.

  C encloses the dipole and therefore the net charge is zero .

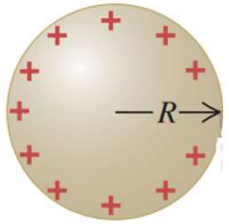
  D does not have any enclosed charge.

# Application of Gauss's law

- Gauss's law is valid for any distribution of charges and any closed surface.
- If we know the charge distribution and if it has symmetry, we can find the electric field by Gauss's law.
- If we know the electric field, we can determine the charge distribution through Gauss's law.
- Note that when excess charge is placed on a solid conductor and is at rest, it will rest on the surface and not in the interior.

# Field of a charged conducting sphere

If positive charges of ***total*** $q$ are placed on a solid conducting sphere of radius R, find the electric field at any point inside as well as outside.




- All charges will reside on the surface and will try to distribute uniformly on the surface. The system is spherically symmetric.

- **Choose a Gaussian surface of radius $r$ centred at the conductor**.
- We choose two cases separately:

$r > R$ means the Gaussian surface will be outside the conductor.

$r < R$ means the Gaussian surface will be inside the conductor.

# Field of a charged conducting sphere outside

For $r > R$ the entire conductor is within the Gaussian surface, so the enclosed charge is $q$. The area of the Gaussian surface is $4\pi r^2$, and $\vec{E}$ is uniform over the surface and perpendicular to it at each point. The flux integral $\oint E_\perp \, dA$ is then just $E(4\pi r^2)$, and

$$E(4\pi r^2) = \frac{q}{\epsilon_0} \quad \text{and}$$

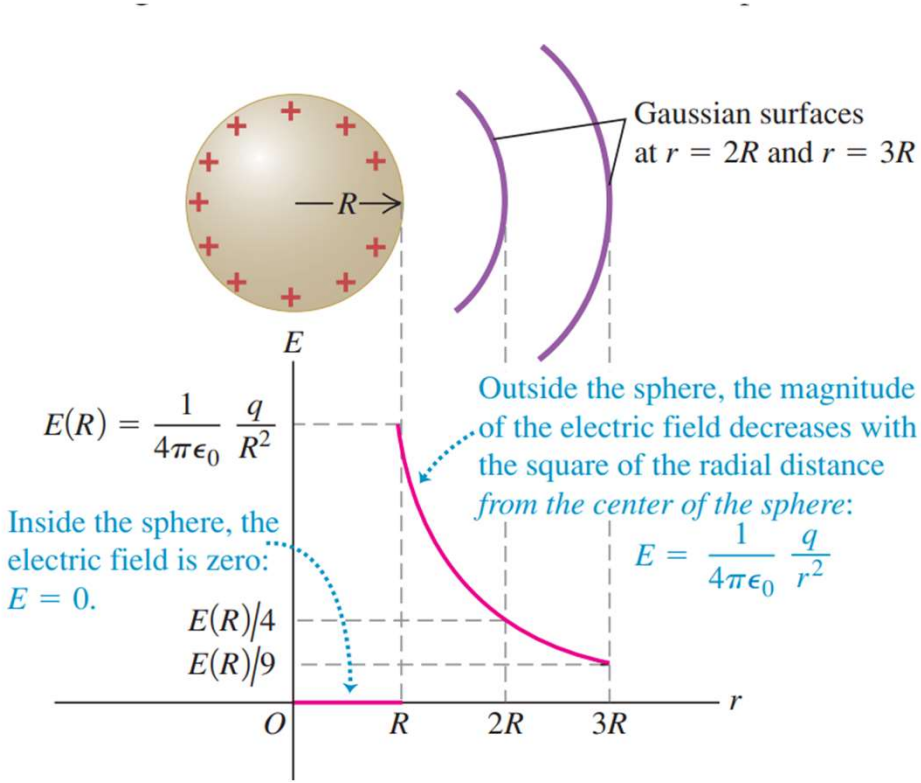
$$E = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2} \qquad \text{(outside a charged conducting sphere)}$$

This expression is the same as that for a point charge; outside the charged sphere, its field is the same as though the entire charge were concentrated at its center. Just outside the surface of the sphere, where $r = R$,

$$E = \frac{1}{4\pi\epsilon_0}\frac{q}{R^2} \quad \text{(at the surface of a charged conducting sphere)}$$

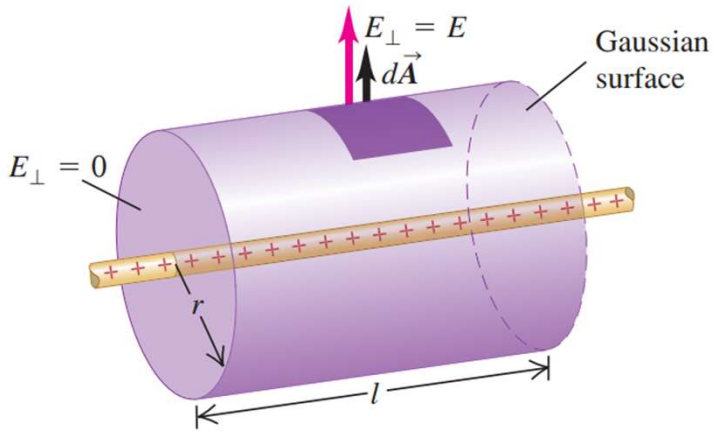
# Field of a charged conducting sphere inside

For $r < R$ we again have $E(4\pi r^2) = Q_{\text{encl}}/\epsilon_0$. But now our Gaussian surface (which lies entirely within the conductor) encloses *no* charge, so $Q_{\text{encl}} = 0$. The electric field inside the conductor is therefore zero.



# Field of a uniform line charge

If electric charge is distributed uniformly along an infinitely long thin wire. The charge per unit length is $\lambda$ (assumed positive). Find the electric field.




- We know that ***E*** is radially outward for an infinitely long wire.
- **Choose a cylindrical Gaussian surface** of radius $r$ and length $l$, coaxial with the wire with its end perpendicular to the wire.

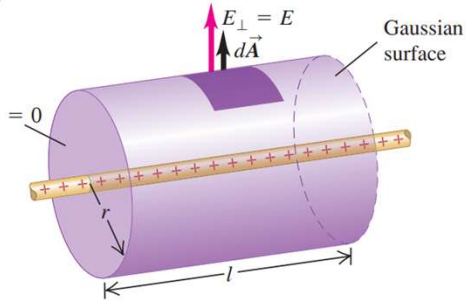
# Field of a uniform line charge

The flux through the flat ends of our Gaussian surface is zero because the radial electric field is parallel to these ends, and so $\vec{E} \cdot \hat{n} = 0$. On the cylindrical part of our surface we have $\vec{E} \cdot \hat{n} = E_\perp = E$ everywhere.

The area of the cylindrical surface is $2\pi rl$, so the flux through it—and hence the total flux $\Phi_E$ through the Gaussian surface—is $EA = 2\pi rlE$. The total enclosed charge is $Q_{\text{encl}} = \lambda l$, and so from Gauss's law,

$$\Phi_E = 2\pi rlE = \frac{\lambda l}{\epsilon_0} \quad \text{and}$$

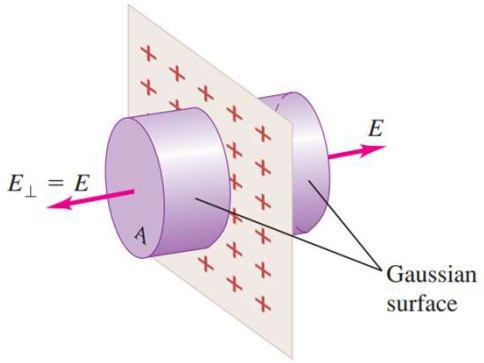
$$E = \frac{1}{2\pi\epsilon_0}\frac{\lambda}{r} \qquad \text{(field of an infinite line of charge)}$$




If $\lambda$ is *negative*, $\vec{E}$ is directed radially inward, and in the above expression for $E$ we must interpret $\lambda$ as the absolute value of the charge per unit length.

# Field of an infinite plane sheet of charge

If electric charge is distributed uniformly on a thin, flat, infinite sheet with charge density $\sigma$ (assumed positive). Find the electric field.



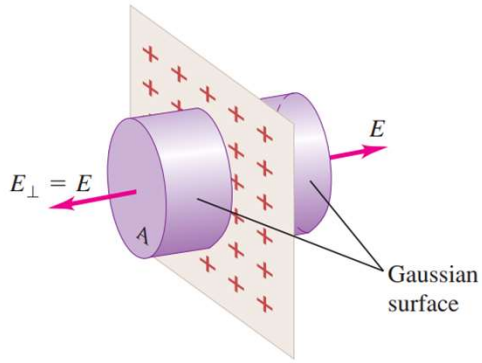

- We know that ***E*** is normal to the sheet and its magnitude is independent of the distance from the sheet.
- **Choose a cylindrical Gaussian surface with ends of Area $A$ with its axis perpendicular to the sheet.**

# Field of an infinite plane sheet of charge

The flux through the cylindrical part of our Gaussian surface is zero because $\vec{E} \cdot \hat{n} = 0$ everywhere. The flux through each flat end of the surface is $+EA$ because $\vec{E} \cdot \hat{n} = E_\perp = E$ everywhere, so the total flux through both ends—and hence the total flux $\Phi_E$ through the Gaussian surface—is $+2EA$. The total enclosed charge is $Q_{\text{encl}} = \sigma A$, and so from Gauss's law,

$$2EA = \frac{\sigma A}{\epsilon_0} \quad \text{and}$$

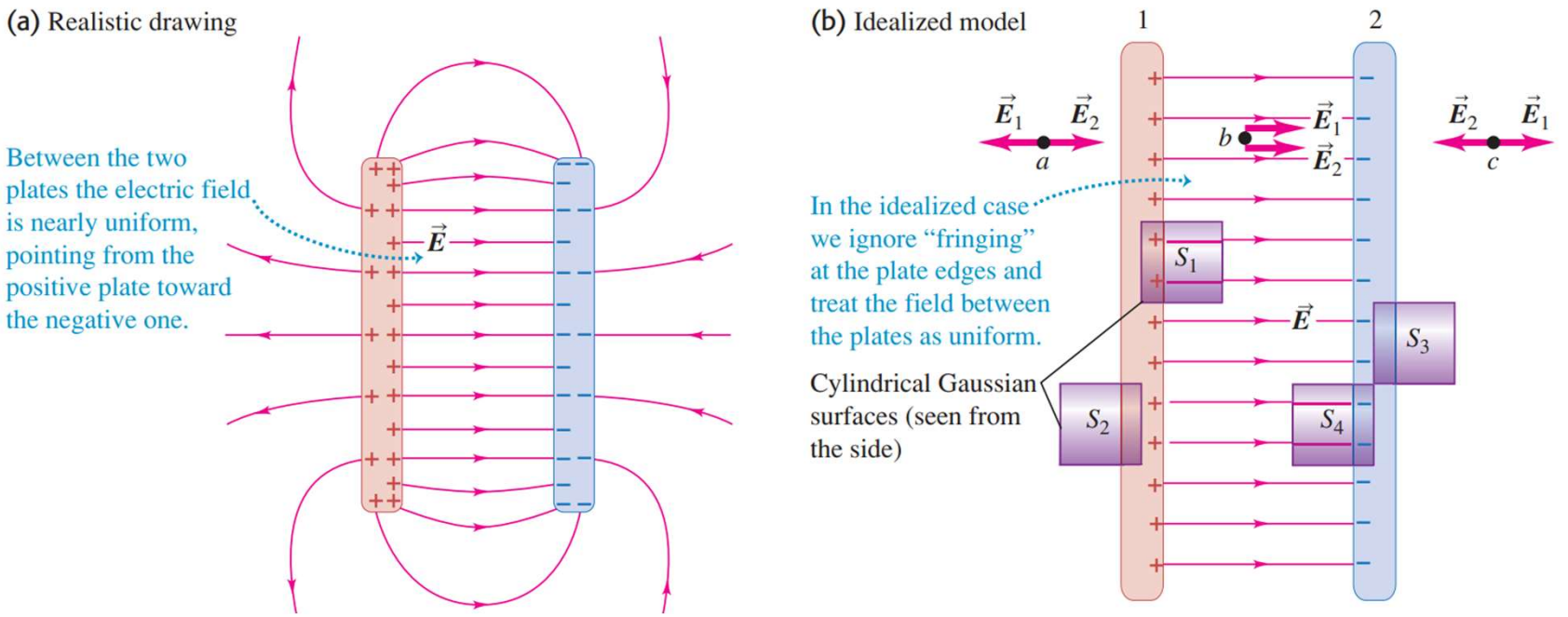
$$E = \frac{\sigma}{2\epsilon_0} \qquad \text{(field of an infinite sheet of charge)}$$




- We know that ***E*** is normal to the sheet and its magnitude is independent of the distance from the sheet.
- **Choose a cylindrical Gaussian surface with ends of Area $A$ with its axis perpendicular to the sheet.**

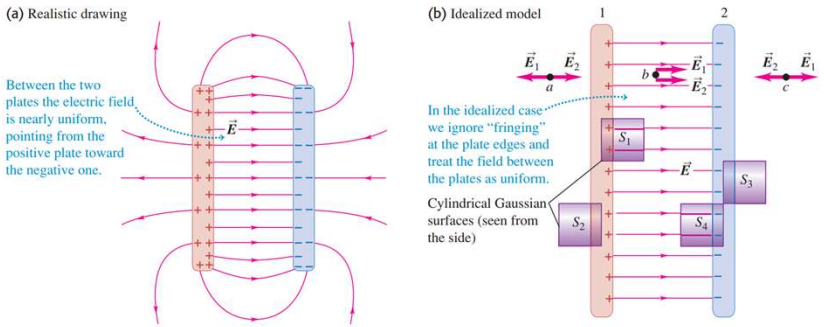
# Example

Two large plane parallel conducting plates are given charges of equal magnitude and opposite sign; the surface charge densities are $+\sigma$ and $-\sigma$. Find the electric field in the region between the plates.



# Example

Two large plane parallel conducting plates are given charges of equal magnitude and opposite sign; the surface charge densities are $+\sigma$ and $-\sigma$. Find the electric field in the region between the plates.







The left-hand end of surface $S_1$ is within the positive plate 1. Since the field is zero within the volume of any solid conductor under electrostatic conditions, there is no electric flux through this end. The electric field between the plates is perpendicular to the right-hand end, so on that end, $E_\perp$ is equal to $E$ and the flux is $EA$; this is positive, since $\vec{E}$ is directed out of the Gaussian surface. There is no flux through the side walls of the cylinder, since these walls are parallel to $\vec{E}$. So the total flux integral in Gauss's law is $EA$. The net charge enclosed by the cylinder is $\sigma A$, so therefore $EA = \sigma A/\epsilon_0$; we then have

$$E = \frac{\sigma}{\epsilon_0} \text{ (field between oppositely charged conducting plates)}$$

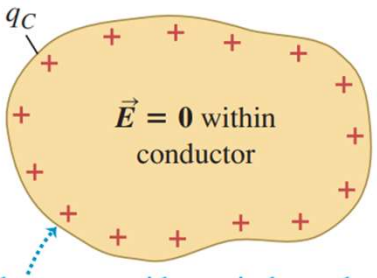
The field is uniform and perpendicular to the plates, and its magnitude is independent of the distance from either plate. The Gaussian surface $S_4$ yields the same result. Surfaces $S_2$ and $S_3$ yield $E = 0$ to the left of plate 1 and to the right of plate 2, respectively.

# Charges on conductors

- In an electrostatic situation the Electric field inside a conductor is zero.

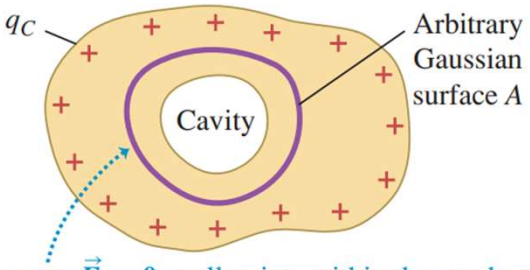
- Any excess charge on a solid conductor will be on the surface.

(a) Solid conductor with charge $q_C$



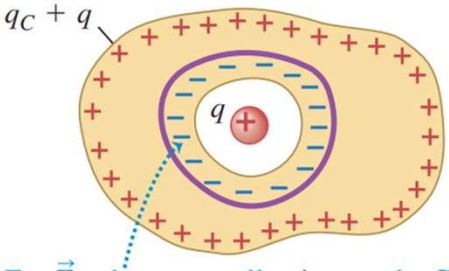

The charge $q_C$ resides entirely on the surface of the conductor. The situation is electrostatic, so $\vec{E} = 0$ within the conductor.

(b) The same conductor with an internal cavity




Because $\vec{E} = 0$ at all points within the conductor, the electric field at all points on the Gaussian surface must be zero.

(c) An isolated charge $q$ placed in the cavity

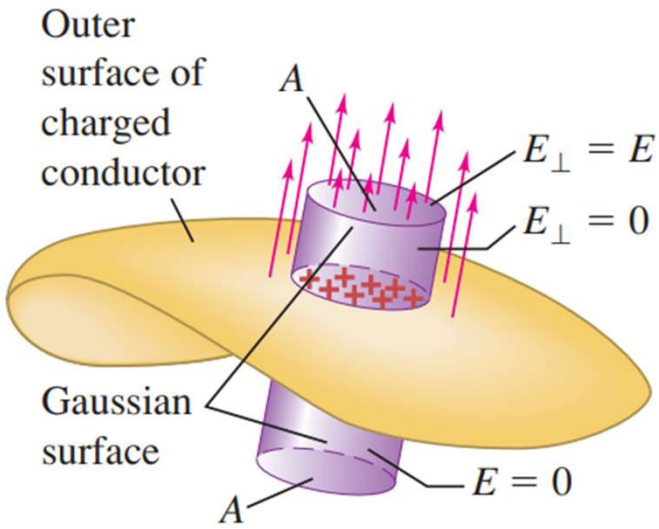



For $\vec{E}$ to be zero at all points on the Gaussian surface, the surface of the cavity must have a total charge $-q$.

# Field at the surface of a conductor

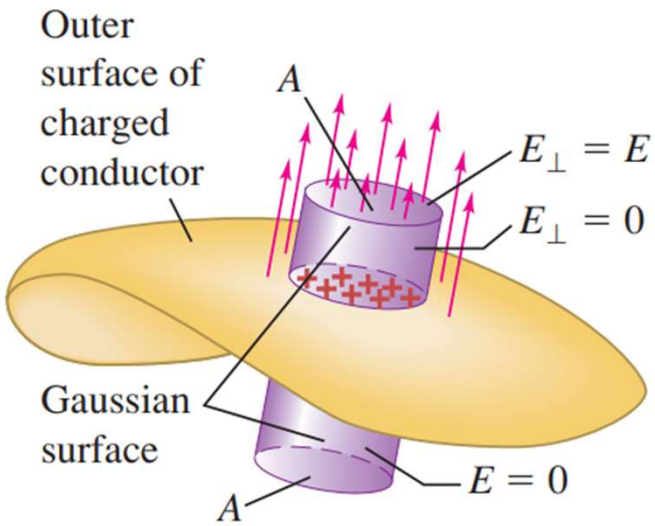
- There is a direct relationship between ***E*** at a point just outside a conductor and the surface charge density $\sigma$ at that point.

- We choose a cylindrical Gaussian surface such that one end lies within the conductor while the other end is outside the conductor.



# Field at the surface of a conductor

- Inside ***E*** is zero at all points.
- Outside the conductor, the perpendicular component to ***E*** is
  - zero on the side walls
  - $E_\perp$ on the end face




- Apply Gauss's law

$$E_\perp A = \frac{\sigma A}{\epsilon_0} \quad \text{and} \quad E_\perp = \frac{\sigma}{\epsilon_0} \quad \text{(field at the surface of a conductor)}$$

# Example

The earth (a conductor) has a net electric charge. The resulting electric field near the surface has an average value of about 150 N/C, directed toward the center of the earth. (a) What is the corresponding surface charge density? (b) What is the *total* surface charge of the earth? ($\epsilon_0 = 8.85 \times 10^{-12}\ C^2/N\,m^2$ and $R_E = 6.38 \times 10^6\ m$)

(a) The direction of the field means that $\sigma$ is negative (corresponding to $\vec{E}$ being directed *into* the surface, so $E_\perp$ is negative). From Eq. (22.10),

$$\sigma = \epsilon_0 E_\perp = (8.85 \times 10^{-12}\ \mathrm{C^2/N \cdot m^2})(-150\ \mathrm{N/C})$$
$$= -1.33 \times 10^{-9}\ \mathrm{C/m^2} = -1.33\ \mathrm{nC/m^2}$$

(b) The earth's surface area is $4\pi R_E^2$, where $R_E = 6.38 \times 10^6$ m is the radius of the earth (see Appendix F). The total charge $Q$ is the product $4\pi R_E^2 \sigma$

$$Q = 4\pi(6.38 \times 10^6\ \mathrm{m})^2(-1.33 \times 10^{-9}\ \mathrm{C/m^2})$$
$$= -6.8 \times 10^5\ \mathrm{C} = -680\ \mathrm{kC}$$

# End of chapter